CellCards 2.0: An Ontology-Driven Platform for Anatomically Contextualized Cell and Cell Line Knowledge Integration

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Abstract

Background. CellCards (https://cellcards.org) is an ontology-driven knowledgebase for standardized characterization of cell types across biomedical domains [1]. Earlier versions represented cell types independently of anatomical context, despite evidence that identical cell types can exhibit distinct molecular and functional profiles across tissues. In CellCards 2.0, we expanded the platform to support anatomically contextualized primary cells and over 30,000 immortalized cell lines represented through the Cell Line Ontology (CLO).

Methods. We redesigned the CellCards semantic architecture to map Cell Ontology (CL) identifiers to multiple UBERON anatomical contexts, enabling tissue-specific disambiguation of primary cells. The platform integrates heterogeneous biomarker and anatomical information from the Human BioMolecular Atlas Program (HuBMAP), including Anatomical Structures, Cell Types, plus Biomarkers (ASCT+B) tables and Functional Tissue Unit (FTU) reference models [2]. To support large-scale cell line integration, we implemented automated semantic ingestion and provenance-aware integration of ontology-linked metadata and cross-references from over 160,000 publicly available Cellosaurus cell line records enriched with PubMed-linked literature metadata. Ontology-related information from CLO and CL was retrieved through Ontobee SPARQL queries to support semantic annotation and hierarchical ontology navigation.

Results. CellCards 2.0 introduces semantically distinct exploration workflows for primary cells and immortalized cell lines. It currently includes over 45,000 cell types or cell line cells. Each cell or cell line cell has its specific web page that includes a list of modules dedicated for the native cell type or cell line cell type. Primary cell interfaces dynamically associate CL terms with tissue-specific anatomical structures, integrated biomarker annotations, Gene Ontology associations, and HuBMAP FTU visualizations [3]. In contrast, cell line interfaces leverage the CLO to organize provenance metadata including species, sex, disease origin, and external database cross-references. The web platform supports ontology-guided search, semantic navigation, and integrated exploration across linked biomedical resources including ATCC, BioGRID, Wikidata, and PubMed-enriched literature. The software demo will showcase tissue-aware cell exploration, ontology-linked biomarker integration, and semantic disambiguation workflows across multi-source biomedical datasets.

Conclusion. By ontologically separating primary cells from immortalized cell lines and standardizing their associated multi-modal datasets, CellCards 2.0 establishes a AI-ready semantic infrastructure for biomedical knowledge integration. This semantic bounding framework reduces the conflation of biologically distinct concepts in downstream knowledge graph and generative AI workflows, improving provenance tracking, semantic consistency, and machine interpretability for biomedical discovery and vaccine research.

Keywords

Cell Ontology; Cell Line Ontology; semantic integration; HuBMAP; biomedical ontologies

References

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